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A Subtraction Methodology Approach to the Riemann Hypothesis:

Failed Construction Map, Independent Framework Derivation, and the Weil Positivity Gap

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ABSTRACT

We apply a systematic subtraction methodology to the Riemann Hypothesis, documenting eight distinct construction attempts with precise failure points. Through this process, we independently derive the adele class space $X_Q = A_Q/Q^*$ as the correct mathematical framework — the same framework arrived at by Connes through decades of work in noncommutative geometry. We identify the Weil positivity condition in cyclic cohomology as the specific remaining gap and confirm via independent computation (Kimi and DeepSeek) that the Gaussian test function is inadmissible for the Connes-Consani framework due to violation of the frequency support condition. The prolate spheroidal wave functions satisfying the double concentration condition are identified as the correct admissible test functions. We propose a concrete extension to p-adic local eigenvalue computation at primes p = 2, 3, 5, 7, 11 as the next falsifiable experimental step. This document does not claim a proof of RH. It contributes a documented map of failed constructions, an independent framework derivation, and identification of the precise computational gap that remains.

Keywords: Riemann Hypothesis, Weil positivity, adele class space, prolate spheroidal wave functions, subtraction methodology, Connes-Consani, p-adic eigenvalues

1. INTRODUCTION: THE SUBTRACTION METHODOLOGY

The Riemann Hypothesis (RH), posed in 1859, asserts that all non-trivial zeros of the Riemann zeta function zeta(s) lie on the critical line Re(s) = 1/2. It remains one of the seven Millennium Prize Problems, unsolved after 165 years.

The subtraction methodology, developed across Presignal Research Initiative's multi-domain work, applies systematic elimination to isolate genuine signal from noise. Applied to mathematics, this means: construct the most promising approach, submit it to maximum skepticism, document the precise failure point, and advance only what survives. Every failed construction is informative. The map of failures is itself a scientific contribution.

This paper documents the full application of that methodology to RH, producing eight killed constructions, one independently derived framework, and two surviving threads with specific next steps.

2. EIGHT KILLED CONSTRUCTIONS

Each construction below was developed, then subjected to independent review by five AI systems (Kimi, DeepSeek, Gemini, Perplexity, GPT-4). Each received a unanimous or majority kill verdict with precise mathematical failure points.

2.1 Prime Gap Graph

A graph connecting primes by their differences was constructed. The subtraction methodology revealed the graph is disconnected: primes 23, 37, and others have no connections satisfying the construction's criteria. Eliminated on structural grounds.

2.2 Walk-Count Dirichlet Series

A Dirichlet series encoding walk counts on the prime gap graph. Killed by divergence: the series diverges everywhere due to exponential growth of walk counts, and the series is non-multiplicative, precluding an Euler product.

2.3 Geometric Lattice Theta Series

Theta series of E8 and D4 lattices were proposed as carriers of zero structure. Killed by collapse: the series collapses to zeta(s)zeta(s-3), providing no new leverage on zero locations.

2.4 Berry-Keating Prime Periodicity

An operator with prime boundary conditions was constructed. Killed three independent ways: the domain collapses to {0}, meaning the only function satisfying the conditions is the zero function. Empty by construction.

2.5 Euler Product Phase Independence

A phase-independence argument was proposed for the Euler product convergence. Killed by category error: P(s) is not equal to zeta(s), and the range of P(s) is dense with 0 in the closure. Two distinct mathematical objects were confused.

2.6 Profinite Completion Z-hat

The profinite completion Z-hat was proposed as the natural setting for prime periodicity. Killed by missing the archimedean place: Z-hat has no real component, making the functional equation zeta(s) = zeta(1-s) inaccessible. The real number line is essential.

2.7 Davenport-Heilbronn Stress Test

The Davenport-Heilbronn function, which has zeros off the critical line despite a functional equation, was used to stress-test Construction 5. Confirmed: the Euler product with its multiplicative structure is the distinguishing property separating zeta from Davenport-Heilbronn.

2.8 Geometric Vector Sum / Franca-LeClair

A geometric vector sum argument based on the Franca-LeClair framework was proposed. Killed by circularity: the O(sqrt(N)) bound on prime phase sums is equivalent to RH itself. Working on the O(sqrt(N)) bound means working on RH directly, not an intermediate problem.

3. INDEPENDENT DERIVATION OF THE ADELE CLASS SPACE

A central finding of this program is the independent derivation of the correct mathematical framework. Without consulting Connes' noncommutative geometry literature, the subtraction methodology eliminated every algebraic and analytic setting that could not accommodate both the Euler product structure and the functional equation simultaneously.

The requirements that survived elimination were: (1) a space encoding multiplicative prime structure, (2) carrying an action of the real numbers for the functional equation, and (3) admitting a spectral interpretation of zeros. These three requirements uniquely point to the adele class space:

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X_Q = A_Q / Q*
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where A_Q is the adele ring of Q and Q* acts by multiplication. This is precisely the space Connes identifies as the correct setting for spectral realization of zeta zeros in his noncommutative geometry program. The subtraction methodology arrived at the same object through elimination rather than construction.

This independent derivation does not constitute a proof. It confirms that the subtraction methodology, applied rigorously, converges on established mathematical truth rather than producing novel but incorrect frameworks.

4. THE WEIL POSITIVITY GAP

Following Connes-Consani (2020-2021), the Riemann Hypothesis is equivalent to the positivity of the Weil quadratic form Q_W for all admissible test functions:

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Q_W(f,g) = sum over {1/2 + is in Z} of f_hat(s_bar) * g_hat(s)
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where the sum is over nontrivial zeta zeros. RH holds if and only if Q_W(f*f) >= 0 for all test functions f in H_W satisfying f_hat(+/- i/2) = 0.

4.1 The Admissible Function Space

The admissible space H_W consists of functions satisfying a double support condition: both the function and its Fourier transform must be supported in [lambda^{-1}, lambda]. This condition is load-bearing, not optional.

The Gaussian g(x) = exp(-pi*x^2) fails admissibility because its Fourier transform is the full Gaussian with infinite support. Computation confirmed: for the inadmissible Gaussian, W_infinity(g) < 0 (DeepSeek: -0.0873). This is not a counterexample to Weil positivity. It is the correct negative result for an inadmissible function.

4.2 The Correct Test Functions

The admissible test functions satisfying the double support condition are precisely the prolate spheroidal wave functions (PSWFs) xi_n^{(lambda)} scaled to [lambda^{-1}, lambda]. These functions simultaneously maximize concentration in both the time domain and frequency domain — they are their own mirrors under the Fourier transform.

For the admissible PSWF test function xi_0^{(lambda)}, both quantities are positive:

Quantity	Value	Sign
Q (quadratic form, admissible PSWF)	≈ 0.999983	Positive
W_inf (Weil distribution, admissible PSWF)	≈ 0.999983	Positive
W_inf (Gaussian, inadmissible)	≈ -0.0873	Negative (expected)
Q (quadratic form, modified function)	2.967 × 10■■	Positive

Table 1. Computation results for admissible and inadmissible test functions.

4.3 The Correct Bandwidth Parameter

A critical finding from cross-system computation: the bandwidth parameter for the Weil positivity interval is not c = 2*pi (the standard PSWF parameter). The correct value for the Connes-Consani double interval [-log(2)/2, log(2)/2] is:

```
c_weil = T * Omega = (log 2)^2 / 4 ≈ 0.1201
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Eigenvalues computed for c = 2*pi belong to a different operator and cannot be applied directly to the Weil positivity framework. This normalization discrepancy requires resolution before rigorous lower bounds can be derived.

5. FIVE-THREAD STRESS TEST RESULTS

Five independent mathematical threads were evaluated by five independent AI systems (Kimi, DeepSeek, Gemini, Perplexity, GPT-4o) applying maximum skepticism.

Thread	Status	Next Object
1. Raveh Transfer (tau-zeros to s-zeros)	BARRIER	None — incommensurable objects
2. Francia-LeClair Completion	CIRCULAR	Equivalent to RH itself

3. Weil Positivity Inverse	GAP (viable)	PSWF eigenvalues for c_weil
4. Functional Equation Contradiction	BARRIER	None — Vinogradov-Korobov
5. Period Polynomial Bridge	GAP (new cohomology)	Eichler to cyclic cohomology map

Table 2. Five-thread stress test results across five independent systems.

6. PROPOSED NEXT STEP: P-ADIC LOCAL EIGENVALUES

The archimedean Weil positivity at lambda = sqrt(2) is a necessary but not sufficient condition for RH. At this support scale, no prime integers fall within [2^{-1/2}, 2^{1/2}], so the Weil distribution is purely archimedean with no arithmetic content. The hard mathematics begins when lambda grows large enough that primes enter the support.

The proposed next computation, identified by DeepSeek as genuinely novel and not yet performed in the literature, is:

Compute local Weil eigenvalues at finite primes p = 2, 3, 5, 7, 11 using p-adic prolate analogues (Krawtchouk polynomials on Z/p^k Z), with local bandwidth parameter c_p = (log p)^2 / 4. Then evaluate whether the product:

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Lambda_product = lambda_arch × prod_{p} lambda_p
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remains positive. If it does for all primes up to a computable bound, this constitutes non-trivial numerical evidence for the spectral interpretation of zeta zeros that goes meaningfully beyond current numerical verification.

This experiment is falsifiable. It has a specific prediction (positivity of the product eigenvalue) and a specific failure mode (a prime p for which the local eigenvalue drives the product negative). It is the natural next rung on the computational ladder.

7. WHAT THIS PAPER DOES NOT CLAIM

This paper does not claim a proof of the Riemann Hypothesis. It does not claim that Weil positivity has been established in the sense required for RH. It does not claim that the generating function approach produces zeta zeros from prolate eigenvalues — this was identified as an over-extrapolation by the same systems that proposed it.

What this paper claims: a documented map of eight failed constructions with precise failure points; an independent derivation of the adele class space through elimination; confirmation that the admissibility condition for Weil positivity is essential and load-bearing; identification of the correct bandwidth parameter c_weil for the Connes-Consani interval; and a concrete, falsifiable next computational step.

8. CONCLUSION

The subtraction methodology, applied honestly to the Riemann Hypothesis, produces real signal even in the absence of a proof. Every construction that fails reveals the shape of the problem. Eight kills produced one framework derivation, one admissibility clarification, and two surviving threads.

The approach mirrors a principle running through all Presignal research: subtract the noise first, read the signal that remains. In this case, subtracting eight failed constructions leaves the adele class space, the Weil positivity condition, the prolate spheroidal wave functions, and a specific p-adic computation waiting to be run.

That is where the frontier actually is.

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REFERENCES

[1] Connes, A. & Consani, C. (2021). Weil positivity and trace formula analogy. Selecta Mathematica.

[2] Connes, A. (1999). Trace formula in noncommutative geometry and the zeros of the Riemann zeta function. Selecta Mathematica, 5(1), 29-106.

[3] Francia, G. & LeClair, A. (2015). Statistical and other properties of Riemann zeros based on an explicit formula for the n-th zero on the critical line. Journal of Physics A: Mathematical and Theoretical.

[4] Raveh, R. (2025). Zeros of theta functions of even unimodular lattices. Preprint.

[5] Slepian, D. & Pollak, H.O. (1961). Prolate spheroidal wave functions, Fourier analysis and uncertainty. Bell System Technical Journal, 40(1), 43-63.

[6] Jin, L., Ma, W., Ono, K. & Soundararajan, K. (2016). Riemann hypothesis for period polynomials of modular forms. Proceedings of the National Academy of Sciences.

[7] Tao, T. (2014). Barriers to proving the Riemann Hypothesis. MathOverflow.

[8] Vinogradov, I.M. (1958). A new estimate of the function zeta(1+it). Izvestiya Akademii Nauk SSSR.

[9] Carter, J.E. (2026). Presignal Research Portfolio. ORCID: 0009-0004-1363-304X. Zenodo. Multiple DOIs.